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P. A. M. Dirac - Session IV

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Interviewed by: Thomas S. Kuhn

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Abstract:

Part of the Archives for the History of Quantum Physics oral history collection, which includes tapes and transcripts of oral history interviews conducted with circa 100 atomic and quantum physicists. Subjects discuss their family backgrounds, how they became interested in physics, their educations, people who influenced them, their careers including social influences on the conditions of research, and the state of atomic, nuclear, and quantum physics during the period in which they worked. Discussions of scientific matters relate to work that was done between approximately 1900 and 1930, with an emphasis on the discovery and interpretations of quantum mechanics in the 1920s. Also prominently mentioned are: Niels Henrik David Bohr, Max Born, Boyland, Louis de Broglie, Johannes Martinus Burgers, Paul Ehrenfest, Ralph Fowler, Peter Fraser, Werner Heisenberg, Ernst Pascual Jordan, Cornelius Lanczos, Edward Arthur Milne, Wolfgang Pauli, David Robertson, Ernest Rutherford, Erwin Schrödinger, John Joseph Thomson, Hermann Weyl; University of Cambridge, Delta Squared V Club, Kapitsa Club, Kbenhavns~Universitet, Merchant Venturer's School in Bristol, University of Bristol, and Universität Göttingen.

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Kuhn:

I'd really like to go back once more to your early times at Cambridge and ask you a little bit more about the people. I wondered particularly, was J.J. Thomson still any sort of a force of importance?

Dirac:

He was still about, yes, yes.

Kuhn:

Did one see anything of him and did he still exert any real influence on the direction

Dirac:

I saw him occasionally in the lab. I think probably Cunningham could answer such questions better than I could. ... I can't remember Thomson's ever giving a colloquium while I was there. He might have given one which I've forgotten completely. He kept a room in the Cavendish and I think he went there occasionally, but not very often.

Kuhn:

I know in one of the few conversations we've had with Professor Bohr he spoke of Thomson's reaction to the Bohr atom and of course then his reaction on the whole subject of isotopes and spoke of him as really having I don't remember his words but they were to the effect that he had really departed from physics at that point. That by his refusal to keep up and his strenuous opposition to what for so many people were critical new ideas, he really ceased to be a member of the profession and I wondered whether there, was any sense of that.

Dirac:

What time was that? What date was that?

Kuhn:

Well, this would have been by 1920, in any case, I think.

Dirac:

I don't remember his coming to any colloquium and taking part in it.

Kuhn:

Can you say any more about Fowler than you already have? Was he not only a person who initiated you, I take it, to quantum mechanics, but was he doing this for many people?

Dirac:

Yes, he was. Yes.

Kuhn:

Was there a real group around him?

Dirac:

There was a group, a small group. The groups then were very much smaller than they are now. But he 'was a great stimulating influence. He was really the center of the quantum theory in Cambridge.

Kuhn:

What form did this stimulus take or how was it exerted? Was it strictly in his lectures?

Dirac:

Well, for one thing he was so excited about it and that excitement was infectious. He went to Copenhagen pretty often and came back and reported at colloquia what he had heard and just generally one caught this excitement from him.

Kuhn:

And this was principally in colloquia and in lectures less, I take it in face-to-face encounters.

Dirac:

I think also then, yes.

Kuhn:

What sort of a person was he?

Dirac:

Well, he was tall and athletic • He was very vigorous and healthy looking and it was quite a surprise when he died so young. Well, there's a picture of him at one of these Solvay Conferences. [showing picture] He was present at this conference here.

Kuhn:

What about Rutherford as a figure here at that point? Clearly he was the leader of major experimental efforts still.

Dirac:

Well, Rutherford dominated the Cavendish. Fowler dominated the theoretical side.

Kuhn:

Did Rutherford have any particular effect on the theoretical side?

Dirac:

I don't think he did, no.

Kuhn:

Did he follow it?

Dirac:

Well, he wasn't much of a mathematician. His mathematics was enough for him to deduce the Rutherford scattering law but not to go further than that. Of course, he appreciated mathematics, but he just didn't tend to follow it himself.

Kuhn:

But he must have heard a good deal from Fowler as to what was going on.

Dirac:

Oh, yes. They were very close.

Kuhn:

Do you have any notion how he responded to these drastic changes that were taking place at the time?

Dirac:

Well, he accepted them anyway. He was mainly concerned with experiments.

Kuhn:

Aside from Eddington, of whom you've already spoken, is there anyone else in the group who had a particular role for the theoreticians or for the experimentalists in sort of shaping the direction of research?

Dirac:

Well, Milne had some role. He was mainly concerned with astrophysics questions about opacity of gases did come into that and of course he was interested in what theory had to say about opacity of gases. So Milne was concerned with it.

Kuhn:

What about people who were students at the time you were studying I don't want to make that too exact a coincidence of dates but —

Dirac:

I don't remember any of them having much influence. One fellow student (at John's) called (Schlapp) who is now professor in Edinburgh, and he had Larmor for his supervisor. Larmor was, of course, interested only in the old classical ideas. He set (Schlapp) to work on classical problems so he was rather away from quantum theory for that reason.

Kuhn:

I realize you've spoken of the accidents involved in your getting Fowler as your supervisor, and your rather hoping you'd have Cunningham.

Dirac:

Yes, but I soon saw that Fowler was more suitable than Cunningham.

Kuhn:

No, I take it that this was at the very least a very fortunate chance which brought you to him; to what extent could a student coming here make a choice himself in the supervisor? To what extent would he be simply given somebody; would that have a more or less permanent effect on the work he did?

Dirac:

It is the degree committee of the board of mathematics which decides who the supervisors are for each research student.

Kuhn:

The student has no choice himself?

Dirac:

Well, he can say he would prefer to work with somebody. I don't think they usually do. I think they just say what subject they want to work on and then the degree committee has to find someone who is willing to work on that subject with the student.

Kuhn:

But they do specify subjects so that at least they would not get off into classical problems if they wanted to do quantum mechanical problems.

Dirac:

They do specify the subject, yes. But usually they don't know very much about what they want to do and they're rather vague about the subject • After all, when a student starts, he doesn't really know what scope there is in different subjects.

Kuhn:

Do you have any notion what proportion of the students at your time were doing classical and what proportion were doing problems that were closely related to the area of quantum mechanics or relativity?

Dirac:

I don't remember. One can look that up by seeing what the well, one can look up the Ph.D. theses of the different students and check up on that. There is a publication giving the summaries of the Ph.D. theses.

Kuhn:

Coming back now to these early papers, the ones before Heisenberg of which we've talked of a good many, there are just a few questions that I still wanted to ask. I think I mentioned in the outline looking through that group of papers I think seven in all of which only the first, the one on dissociation under a temperature gradient, doesn't quite fit the pattern I wanted to point to. All of the other six papers have in common something which I don't see in the same way in your later work; that is, in each of these cases in one way or another you're taking a topic that's relatively standard in the literature and criticizing existing results and doing them, putting them on a firmer basis than before.

Dirac:

Well, that's how I got subjects for research. I was reading quite a bit, and well, every now and again I saw a chance of improving on something which I had read and that led to a paper. Of course, it was really the same with Heisenberg's first paper. I saw a chance for improving on that, by concentrating on the non-commutation.

Kuhn:

I suppose so, but the whole feeling of that paper is very different —.

Dirac:

You can say that it was all built up from. Heisenberg's paper in the same way the previous ones were built up from other papers.

Kuhn:

I guess I could; I wouldn't, in the sense that this goes, in a way, so much further —.

Dirac:

But I started out in the same way. Let's put it like that.

Kuhn:

Was that a sort of work that you found very satisfying or did you view these as parts of education?

Dirac:

Well, it was really the only kind of work I knew at that time. Of course, the difference with that first paper, “Dissociation under a Temperature Gradient” was that I hadn’t done any reading before that. That was the first subject that was set to me while the later ones did depend on my reading.

Kuhn:

You had a larger role in finding them.

Dirac:

Yes.

Kuhn:

I started, I realize, two or three times to ask you the question and I’ve never finished the sentence for some reason or other but in the paper on “The Doppler Effect [Principle] and the Bohr [Frequency] Condition” [No. 3] in which you produce a relativistically invariant formulation by treating the energy change from the phase function —

Dirac:

Yes.

Kuhn:

I’ve brought this up before in connection with de Broglie’s work because he does and I can’t remember now whether he does it in the piece in the Phil. Mag. or not or whether he does it just a little bit later. He does things very much like that four dimensional formulation of yours, not for the same problems but in the same area. I think he does the identical thing with the phase function. I wondered whether there was any likelihood that you had seen things in his work that were carried over there or whether that idea started from scratch.

Dirac:

I just don’t remember. Was de Broglie’s paper published sometime before mine?

Kuhn:

It’s published nearly at the same time.

Dirac:

One would have to check up on the dates.

Kuhn:

It's only very late, in the late stages before the thesis that de Broglie in his publications begins to do this sort of four dimensional formulation.

Dirac:

Yes, I don't remember being influenced by de Broglie in writing that little paper.

Kuhn:

I'm also curious in looking at those early papers as to what determined the place one would choose to publish a paper. Those papers come out variously in the Philosophical Transactions, the Cambridge Philosophical Society Proceedings one of them in the Phil. Mag.

Dirac:

There were none of them in the Transactions. I think the more important ones were sent to the Royal Society Proceedings. I think Fowler suggested where I should send them to.

Kuhn:

What besides the importance would be a determining factor?

Dirac:

A very short note would hardly be a suitable paper for the Royal Society [Proceedings].

Kuhn:

But what about there's the Phil. Mag., Monthly Notices of the Royal Astronomical Society, and the Proceedings of the Cambridge Philosophical Society.

Dirac:

Well, the Astronomical Society would only take papers which were of astronomical interest. They are not general in the way the others are.

Kuhn:

What about the Proceedings of the Cambridge Philosophical? Was there any sense that these were likely to get lost for a larger audience? I realize I have very little notion how widely that journal was read.

Dirac:

It was read pretty widely, yes. It has quite a high standing. It certainly did at that time, perhaps not quite so much now because the *cal Review* now dominates the situation. But it did have quite a wide standing.

Kuhn:

So one didn't have the sense that something placed there was likely to be lost.

Dirac:

No, oh, no. It certainly wouldn't be lost. I think all the libraries have it. I haven't come across any physics library which doesn't have it.

Kuhn:

No, I think that would be quite true, except perhaps for the library in Rome, which I discovered was in this period lacking in several journals, including *Naturwissenschaften*.

Dirac:

Yes.

Kuhn:

But I think you're quite right that it would be in many libraries. Of course, I also notice that one can't count in this period on any given physicist's having followed all of the journals that the library would have.

Dirac:

Yes.

Kuhn:

You would almost certainly have followed the Royal Society and possibly the *Phil. Mag.*

Dirac:

It was of higher standing to get a paper published by the Royal Society, but in any case it would be widely read.

Kuhn:

When papers come in either to the Royal Society or to the Phil. Society here they're always communicated by a member.

Dirac:

By a Fellow of the Society, yes.

Kuhn:

Is it permissible to assume that the person who has communicated the paper has really been over it fairly carefully? Does he take any real responsibility for it?

Dirac:

He should take some responsibility for it. I would, if I were sent one. I wouldn't follow all the details, but in any case I'd see that it wasn't a crank. I'd see if the main ideas are worth writing about, even if I don't check to see if the conclusions are all correct.

Kuhn:

Do you think you were perhaps better than most about this? I ask this only because there are some interesting patterns, I asked Louis de Broglie about such endorsements and he immediately assured me that so far as transmission for publication in the Comptes Rendus was concerned, almost nobody really looked at those papers or at the papers he communicated if they came from, or through, somebody he knew. There was apparently no reason for presumption of some real responsibility toward the subject matter being taken by the person who presented the paper. Was it as loose as that in England?

Dirac:

Well, the person who communicates the paper does have some responsibility, but there is also a referee. He doesn't have the sole responsibility.

Kuhn:

Was that true for all of the journals; that is, true for the Royal Society also for the-

Dirac:

Cambridge Society. But for the Philosophical Magazine anyone can communicate a paper. I mean the author can send in his own paper without having to get someone to communicate it.

Kuhn:

But there will always be a referee in addition to the person that communicates it for all of these.

Dirac:

Yes.

Kuhn:

We've talked scarcely at all about the paper on detailed balancing. I'm a little unclear as to what to ask about it except that it's a particularly interesting paper. It is the first one that gets sent to the Royal Society. How did you come on that problem?

Dirac:

Well, Fowler was interested in this question of detailed balancing. It was an important question of statistical mechanics at that time. People had thought of this, I don't remember who thought of it first and it did enable one to get information.

Kuhn:

I think probably Klein and Rosseland, are the first people who at least show it in the sort of application.

Dirac:

Well, Fowler had taken it up and was using it,

Kuhn:

How much did he work with you on that? Was he a more active participant in that paper than in the others?

Dirac:

I think I mainly worked it out by myself and then showed him a more or less finished version.

Kuhn:

Were there any particular reactions to it or follow-ups on it?

Dirac:

I don't remember any.

Kuhn:

Or with respect to the pair of papers on the adiabatic principle that you do? One of them I think for the Royal Society and the other one in the Cambridge Proceedings.

Dirac:

Yes, I've forgotten that one in the Cambridge Proceedings.

Kuhn:

It's a very short one that deals particularly with the case of magnetic fields, velocity dependent potentials. It discusses the treatment, corrects and justifies the result, among other things, of a way of treating magnetic fields that Sommerfeld had used in Atombau. And those, I take it, do come out of your reading.

Dirac:

Yes.

Kuhn:

I'm particularly interested in the first of those papers in which you develop a more general and more powerful technique for evaluating invariants through points of frequency degeneracy

Dirac:

Is this the one in the Royal Society?

Kuhn:

Yes. And there you build it entirely, so far as your citations are concerned on one early paper of Burgers.

Dirac:

Yes.

Kuhn:

In the Proceedings of the Amsterdam Academy, I think. My impression is, although I don't know that literature very well, that there'd been a good deal else that had happened in the interim.

Dirac:

Some other people have written about it, I remember that someone writing a summary of it for one of these magazines' which give summaries of scientific papers, had written unfavorably about my paper and said that some other paper was better and no one would read mine. I thought that was an unfair comment on my paper and the person who wrote that summary hadn't appreciated the difference between mine and the other work, namely that in my paper one could check whether the conditions are fulfilled or not without having to integrate the adiabatic equations, while in the other paper one would have to integrate the adiabatic equations. It was the only time I have been dissatisfied with the comments which someone has made on my work,

Kuhn:

Had you known the other paper? The one that he felt was better —.

Dirac:

I don't think it came out before mine. I think it was independent. I'm not sure of that.

Kuhn:

Well, if it was independent and didn't come out before yours, it's a rather odd sort of criticism.

Dirac:

Well, he just said the other paper was better and that the reader needn't bother to read my paper.

Kuhn:

Did you or anyone else apply the technique you there developed?

Dirac:

No. Well, we very soon lost interest in it because of the appearance of the new theory. If a new theory hadn't come out we would probably have worked more on it.

Kuhn:

In that paper there is a very intriguing remark which I've never tried to follow up in my own mind to see how it would work, that there is a sense in which you can reduce all problems involving multiply periodic systems to the two-dimensional case, because of the selection rule.

Dirac:

I don't remember that remark. Might be wrong.

Kuhn:

I'm not sure whether I can say enough more about to bring it back.

Dirac:

Well, it's best to look up the paper then. [Begins to search for paper]

Kuhn:

I didn't mean to be challenging the accuracy of what was said, nor was I indicating skepticism about the result but I really wanted to ask - I would say this is the sort of result appearing out of the mathematics that Ehrenfest, for example, who thought very often of adiabatic invariance as a key to the finding of a new quantum theory, I think would have jumped on. For him it would have been something that immediately indicates an unexpected characteristic of multiply periodic systems, that may tell us something about Nature, rather than just about the mathematics. I wondered whether you had any such sense yourself. That is, it is rather odd at least that you discover, doing this problem with a technique usable only for two dimensions that somehow or other the multiply periodic system acts two-dimensionally because of selection principles. Yet the selection principles come at this point not from the mathematics - I mean not in the sense that later they come from group theory - but from a variety of rather elaborate

physical arguments. That the method of this special characteristic of the mathematics suddenly fits to the selection principles to make the whole problem manageable as a two-dimensional problem, would have caused some people to be just delighted and to think, "I've got something really fundamental here."

Dirac:

I don't remember the details of the work. Is the work correct?

Kuhn:

I have not been able to study any of these papers closely, but at least it makes good sense.

Dirac:

Yes.

Kuhn:

I think certain of the people who were doing this sort of work at the time, might have been disposed to try to make a great deal out of that.

Dirac:

Well, I didn't feel it was very important. I felt the whole of this work was really very feeble and that we hadn't got the right clue.

Kuhn:

But as far as you could remember this didn't seem to you to be very likely in the nature of a clue that could point to something deeper.

Dirac:

No, I would probably have gone on to work on it if it had..

Kuhn:

Did you have any of the feeling about adiabatic invariance that Ehrenfest I think in particular did have. I mean this was for him something to be pursued as a clue to 'quantum mechanics,' In fact, the term seems to originate around Ehrenfest. I first see it in Burgers' thesis. Born, I think, feels that he was the first to use it but that's much later

and you find “Quanten Mechanik,” in quotation marks, beginning to figure as early as 1918 around Ehrenfest with the notion that adiabatic invariance somehow point to an underlying regularity in Nature which is closely associated with the heart of the new theory, whatever that will be.

Dirac:

Well, people did feel that it was a useful principle. That and the Correspondence Principle was all one had to work on in those days. But still it was feeble because it applied only to very special conditions.

Kuhn:

What about the Correspondence Principle? Did that seem something very fundamental or did it seem to be a crutch?

Dirac:

It always seemed to me to be a bit vague. It wasn't something which you could formulate by an equation.

Kuhn:

Yes, I suppose the answer is just that you couldn't write it as an equation. After you had written your equation you would know what the Correspondence Principle said should happen to it, but you'd have to have the equation first.

Dirac:

All it said was there was some similarity between the equations of quantum theory and the equations of classical theory. I don't believe it was more definite than that.

Kuhn:

Well, it said that the Correspondence should be manifest as h goes to zero, and in some formulations that it should reduce to an identity, as h goes to zero. Of course, I think that in the earlier forms where it's applied initially only to frequencies but then also to frequencies and amplitudes, that you've said that the equations should become the same for large quantum numbers or for small h .

Dirac:

People held onto the Correspondence Principle and the adiabatic principle because it

was the only thing they had to hold onto.

Kuhn:

Well, my feeling is that in Copenhagen in particular the attitude toward the Correspondence Principle would have been that it was something more than a heuristic tool in the sense that you're suggesting. Is that wrong? It may well be.

Dirac:

I'd say that what the Copenhagen people think is correct by definition, and probably I didn't fully appreciate the Correspondence Principle. I expect that's the answer • I didn't fully appreciate it because it didn't have the kind of precision which I like to have.

Kuhn:

Do you suspect that Fowler felt much the same way about it?

Dirac:

No, I think he was quite happy with it. The adiabatic principle was more definite to me than the Correspondence Principle because it did give some precise equations.

Kuhn:

I ask then just once more in this period before the Heisenberg paper appeared. Do you remember concretely other problems that you did which didn't come to publications I can give a hint on one and I don't know what it means, but there is a letter from Fowler to Bohr in Copenhagen in which Fowler among other things transmits a memorandum or something of the sort of yours, and so far as I know that no longer exists, which deals with a problem that he simply refers to as Wentzel's phase. I don't know what Wentzel's phase is. Undoubtedly if I read all the Wentzel papers from these years I might, but there's at least nothing in the titles

Dirac:

You'd better ask Wentzel about that.

Kuhn:

Well, indeed I shall ask him but he won't be able to tell me

Dirac:

Why shouldn't he be able to?

Kuhn:

He may be able to tell me what Wentzel's phase is, he won't be able to tell me why you were interested in it. That is, I raise this not to find out what Wentzel's phase was, which we can do any one of a number of ways, but to see whether it brings back any more detail to you about that problem or other problems you may have been working on that one can't get at now from the published literature.

Dirac:

Well, I would have to refresh my mind about what the Wentzel phase is. I might remember something. I know I worked quite a bit on the Weyl's electro dynamics and that's not connected with quantum theory at all.

Kuhn:

Did you think it might supply any hints toward the quantum problem?

Dirac:

No, I thought it might be a correct description of Nature and I was quite disappointed to find that it wouldn't work. I made a detailed study of the field around an electron and I found that the equations would really require that the charge is not constant but varies with time by a certain coefficient depending on the mass, Or is it the other way around the mass isn't constant and varies with a coefficient depending on the charge. One or the other of the two possibilities.

Kuhn:

You didn't publish that?

Dirac:

No, I didn't publish it. But it meant that the Weyl theory is untenable, and everyone seemed to agree that the Weyl theory was untenable although for different reasons.

Kuhn:

Do you have recollections of any other things of this sort?

Dirac:

Well, I was working a great deal on action and angle variables. That was my main subject of work.

Kuhn:

What sort of things do you think you were trying to do with them?

Dirac:

I suppose the main problem was to try and introduce them or something corresponding to them, for systems which are not multiply periodic - the helium atom in particular. Trying to extend dynamical theory to have something to play the role of these variables for systems which are not multiply periodic. It seemed to me at that time that that was the only way in which one could develop quantum theory.

Kuhn:

If I might move now to your introduction of the distinction between c numbers and q -numbers in the first of the papers that does an application to the hydrogen atom. [Paper No. 9] There had been no hint of something of this sort in the first paper. [Paper No. 8] In formulating the two sorts of numbers one has to find one which has the algebraic properties of the quantum mechanical entities, the other which can be represented by algebraic numbers, and then some way of correlating these two. This is a terribly big step and very clearly formulated in the paper. How did you come upon the distinction?

Dirac:

Well, that was the whole difference between classical theory and quantum theory. You might ask why I was able to manipulate the symbols so easily. I think I had gotten used to symbolic methods earlier. The (stress diagrams) which engineers use is something like symbolic methods for getting results. And in projective geometry some of the methods tend to approach symbolic methods.

Kuhn:

I've never done any projective geometry and I hadn't realized that they were so often symbolically formulated.

Dirac:

Well, I think someone called Grassmann invented a symbolism for working out things in projective geometry.

Kuhn:

Grassmann?

Dirac:

I'm not quite sure that I've got the name right.

Kuhn:

In any case there was a book with which you worked at one point that at least made some reference to this sort of treatment.

Dirac:

Yes. Fraser's lectures were built very much on one to one correspondence deducing results simply by working from one to one correspondences. It's a very powerful, sort of a magic way of getting results. That sort of training was very suitable for later on developing symbolic methods.

Kuhn:

Does the one to one technique that you think of their bear some resemblance also to the way you develop an algebra for q-numbers?

Dirac:

No, not in detail.

Kuhn:

You speak in this paper of the correspondence between q-numbers having an analogy to c numbers and to classical observable quantity. You say there are occasions on which more than one q number will be available as the analogue for a single c-number and then you have to make a decision as to which is the more important, and give as an illustration of this the fact that you can develop for frequency in q number notation things that I would want to say are the analogue on the one hand to classical orbit frequency and on the other hand to classical transition frequency; I wouldn't want to say that there are two q-numbers which are equivalent, which are analyzed for the same c number, I've not quite understood really what that problem was for you, I mean what a more proper one to one correspondence would have been. I think there's clearly something that was bothering you.

Dirac:

Probably the idea was simply wrong. I don't remember this about the frequencies.

Kuhn:

This one I would be grateful if we could have a look at.

Dirac:

That's a question of naming the frequencies in the quantum theory.

Kuhn:

I take it that that phraseology indicates that you would really have been better satisfied, or felt that the situation was neater, if in practice there had been only one thing that showed up as a frequency that this is an aspect of ambiguity which has to be eliminated by intelligence that goes beyond any directions you are able to give.

Dirac:

Yes, I would agree with that. It's largely a question of terminology.

Kuhn:

But I'm perplexed as to why, in the light of the whole structure of this theory, it comes as a disappointment to have two things —

Dirac:

The correspondence between the classical and quantum theory is not quite so close as one previously expected.

Kuhn:

But isn't it exactly that this is because there is the q-number formulas are all modeled on this classical formulas. In the classical formulas you do get two sorts of frequencies, the orbital and the transition. Here again you get them again and it seems to be an exact, it seems to be an example of the analogy rather than a break-down of the analogy.

Dirac:

Well, it means a lack of one to one correspondence between quantum and classical

theory.

Kuhn:

Classical in the sense of pre-quantum mechanical classical not in the sense of Bohr classical.

Dirac:

Yes, pre-quantum. Yes, it is classical in that sense, and it is a failure of one to one correspondence between quantum and classical theory.

Kuhn:

Classical in the sense of pre-quantum mechanical classical not in the sense of Bohr classical.

Dirac:

Yes, pre-quantum. Yes, it is classical in that sense, and it is a failure of one to one correspondence between the two theories.

Kuhn:

Yes, I guess this is what I was missing, that you shouldn't have had such things as both transition frequencies and orbital frequencies.

Dirac:

Classical here does mean before Bohr orbits.

Kuhn:

Before Bohr orbits you only had orbital frequencies and by the same token as you come around again you should only have one sort.

Dirac:

Yes.

Kuhn:

This brings me to your paper on the Compton effect and one of the problems I want to

raise here is one that I think you may want to talk about and I'd certainly be very grateful, not only with respect to this problem but also with respect to a good deal that happens later, if it leads to that. In the first paper since before you have gotten into quantum mechanics, in the Heisenberg sense in which you attempt to relativism the equations, had you felt from the beginning that the fact that they were non-relativistic equations was a limitation?

Dirac:

Which paper are you talking about now?

Kuhn:

This is the paper in which you introduce a four dimensional formula, in which you introduce time and energy as conjugate variables. (Paper No.11)

Dirac:

The one on the Compton scattering, yes. Well, of course, I knew from the beginning that the theory ought to be relativistic. Relativity was very well established by then.

Kuhn:

When you were working on action and angle variables, had you attempted to do that in a relativistic formulation? Had you had any notion that there might be then a clue?

Dirac:

I can't remember doing that. I don't think so, because the main problem there was the helium atom and relativity doesn't come in when you've got a heavy nucleus

Kuhn:

What I'm in part here groping towards I think there were a good many rather different attitudes in the profession as to what sort of validity was left to a non-relativistic approach in view of the existence of relativity theory.

Dirac:

Well, it was certainly a very good approximation when you've got an atomic nucleus present. I think my engineering training was a help in reconciling that kind of an approximation which is so obviously sensible from the practical point of view,

Kuhn:

In the forward to the third edition of your Mechanics you speak of one very important change in this edition being the sense in which you've now adopted the non-relativistic state formulation.

Dirac:

Yes, I remember making that change.

Kuhn:

But you had thought at one time that the quantum mechanics should in its neatest and best form be a relativistic quantum mechanics and that there is some sense here that there are things that we cannot preserve in relativistic mechanics that seem very natural to quantum mechanics and that we only get in a non-relativistic form.

Dirac:

Well, when I wrote the first edition of the book, I tried to present the theory in terms of basic ideas which are relativistic and I was lecturing on the subject every year to students and I found that it just wasn't practical in a first presentation of the subject to people to whom it is quite new. You want to present the basic ideas in the simplest possible way and for that purpose it is more convenient to use non-relativistic concepts.

Kuhn:

But surely what you say in the preface to the third edition goes deeper than the question of pedagogic practicality. It seems to me that it winds up with the remark that there is a problem that perhaps we all ought to take more seriously.

Dirac:

I don't remember just what I said. I think it still applies to the present day. People who write textbooks on field theory try to formulate everything in terms of relativistic concepts and I think that they make the theory more obscure than it need be by doing so.

Kuhn:

But do you think this is really a problem in obscurity of presentation, not of there being any sort of more fundamental miss-match between a relativistic formulation and a fully adequate quantum theory?

Dirac:

I think there is a miss-match and I think it's not yet solved.

Kuhn:

I take it that this preface does hint at that, or perhaps more strongly than hints.

Dirac:

Well, maybe it does. I've had the idea a long time that there is some basic discrepancy here. I didn't know it went back as far as that.

Kuhn:

What I'd like to do is to get you to talk about that, your early and later experiences in trying to keep quantum mechanics relativistic.

Dirac:

Well, I only know about much later experiences. I have some difficulty in remembering the beginnings of these ideas.

Kuhn:

Well at what point would you feel fairly sure of saying "I was really beginning to think this problem might be fundamental"?

Dirac:

Well, just during the last few years in some of my lectures I've been saying that I think the four-dimensional picture of the world is not the complete answer. I don't know whether you are interested in these very late developments. It's something which has been growing on me for a very long time.

Kuhn:

To what extent does your reason for saying that the four-dimensional formulation is not the final word grow out of problems of the four-dimensional formulation by itself and to what extent does it grow out of the problem of matching it to quantum mechanics? That obviously doesn't cut a nice logical line between two sets of problems, but —.

Dirac:

I expect that it began through my finding out that it is difficult to teach students if you work with the four-dimensional picture all the time. That seems to suggest that the four-dimensional picture is not really so good as it might be. If it was really the final word this difficulty should not occur. And all the vacuum fluctuations that you get with field theory contribute to this point of view.

Kuhn:

But you think then that this sense of miss-match is really very likely recent history, that you were into the middle thirties, say, pretty well content that the problems that remained in matching quantum mechanics and relativity were problems of detail that would surrender to

Dirac:

Well, I thought that there were difficulties because of the infinities in the field theory. That occurred pretty early, I think already before 1930. It was a disappointment because the first few years one thought that the quantum theory of Heisenberg would be the answer to everything. Then as soon as one began to develop field theory, one saw that it wasn't.

Kuhn:

I had hoped that there might be roots of this problem of relating the two that went as far back as your earliest introduction of relativistic formulation. [Paper No. 1] It is the one in which you introduce W and t as conjugate variables in a Hamiltonian formulation. But that's entirely new to me when I see it there but I'm not at all clear as to whether there had been other formulations of that sort before. I mean in the classical Hamiltonian equation with W and t occurring as conjugate variables.

Dirac:

I think it has occurred in classical theory but I'm not quite sure just where.

Kuhn:

You develop it classically first and then simply apply commutation relations to W and t ; the classical formulation is one I hadn't seen. I don't think it's in Whittaker, for example.

Dirac:

I think it is rather standard that you can count time as an extra variable and introduce something conjugate to it.

Kuhn:

Do you think it was relatively standard at the time? I don't know of another place where this point had been put previously in this way, but I'm not at all sure it hadn't.

Dirac:

Well, I think I might answer you in much the same way that when I wrote that I felt that probably it had been done before, but it was less trouble to me to present it as something new than to search for a reference. A good deal of my work was like that • It happened rather often that there was something which I thought was very likely done before, but seemed to be a great nuisance to look through all the references to try to find it, and if it doesn't take much trouble to publish it, one could publish it again without claiming either that it's new or that it has been done before. For example the Hamiltonian theory for an electron moving in a given external field. I think that's probably been done before although I haven't found the reference to it. Just the classical equation, working with the Hamiltonian which is quadratic in δ as well as quadratic in the three momentum variables.

Kuhn:

My guess is that it's also been done before, but I'm not quite certain.

Dirac:

It quite possibly has been done around 1910 or something like that. Maybe Cunningham could answer that. I suppose I was lazy with regard to looking up references.

Kuhn:

To what extent, if you knew you had seen a point of this sort somewhere else, but couldn't remember where, would you have tried to find it or have mentioned it as "someone has shown" or something of this sort. Because I take it when you mean that you didn't look it up, you meant that you didn't bother to go back and check all the literature to see whether somebody had in fact done this before. That so far as you knew, nobody had done it but -.

Dirac:

Well, I think with regard to using W and t as conjugate variables in the classical theory, I did feel that I had seen it somewhere before but I couldn't remember where and it might even have been mentioned in a lecture. I think I had seen it before; it's not my own

invention. It's probably in Whittaker or —.

Kuhn:

In this first Compton effect paper, you treat the field as a classical variable with no attempt at this point to quantize it.

Dirac:

That is so, yes.

Kuhn:

Did you think of that as being an approximation at that point? Would a fuller theory have introduced the field also as q-numbers? [Brief interruption]

Dirac:

I don't think I felt conscious of it as an approximation. So often in one's engineering work one makes approximations which one isn't conscious of; one treats a body as rigid although actually it has elasticity in it and it gives; one just isn't conscious of these approximations.

Kuhn:

Well, Born and Jordan, and then again Heisenberg, Born and Jordan, had both done something treating the fields as quantum mechanical quantities as quantized. This is really after that.

Dirac:

Yes, after that, not at that stage.

Kuhn:

I wondered whether from the beginning you had taken it for granted that in a full treatment the magnetic and scalar potential would also come out as quantum mechanical quantities, would also satisfy some sort of commutation relations.

Dirac:

I can't remember whether I thought of that at the time I wrote the quantum paper. Probably not.

Kuhn:

Some people seemed to have the feeling that this whole attempt to quantize the field, at the very start, was nonsense and that there was nothing that had happened yet that made it necessary to break with electromagnetic theory in those.

Dirac:

Well, one had the treatment of the hydrogen atom where the field is certainly not quantized, it's just the Coulomb field. That worked very well. When I wrote that Compton paper I expect I felt the same way about the field in the work on the hydrogen atom.

Kuhn:

Do you remember reacting at all to those late sections of Born-Heisenberg-Jordan work in which they almost from the start, and I think this was largely Jordan's doing, pushed into treating the field in matrix terms also?

Dirac:

I think that was later, I don't remember just when it came out.

Kuhn:

There's field quantization, of a sort, in the immediate follow up on the Heisenberg paper.

Dirac:

Yes.

Kuhn:

So I think you'd have seen those papers by then. Let's come to the Schrödinger paper • Did you know of that work, the Schrödinger wave equation immediately?

Dirac:

People told me about it, but I think I was rather delayed somewhat in reading it just because I had a good method of my own • If you've got one good method and can devote your whole attention to it, you don't really want to be diverted onto another method. So at first I did not pay attention to it, but I did later on because I was told it

was so important.

Kuhn:

How did you feel about it when you saw it?

Dirac:

I suppose one does feel a bit annoyed if one has one method which works perfectly well and then one finds one has to learn about another method also, and fit the two together.

Kuhn:

Did you suppose from the start that they were presumably equivalent? Supposing that you did not think that they were equivalent at the start. Did this mean that presumably Schrödinger's was just wrong? Or that one or the other was wrong?

Dirac:

Probably my first reaction of all would be to think that Schrödinger was wrong, because it was so different from another method which one felt pretty sure was right. I think that was my first reaction.

Kuhn:

How about the manner in which he interpreted the equation —.

Dirac:

How did he interpret it in the first place?

Kuhn:

Well, that is a real wave. He supposes, for example, that radiation is perhaps due to interference effects between waves of different frequencies. The frequencies of these waves are the orbital frequencies and there will therefore be an interference effect that's the difference between two orbital frequencies.

Dirac:

The question as to whether the waves are real or not would not be a question which would bother me because I would think upon that as metaphysics.

Kuhn:

He would also think that the wave packet which one decides ultimately tells you probability density is in fact the electron.

Dirac:

The wave packet didn't come in until quite a bit later. I don't think that it did.

Kuhn:

Quite early Schrödinger is using something rather like a wave packet - certainly in the problem of spreading. He talks about dimensions of the electron, conceived as the spreading of the wave function, as compared with atomic dimensions, very nearly at the beginning of that work.

Dirac:

Well, for people who want to interpret the waves as real, the spreading of the wave packet does present a serious difficulty.

Kuhn:

But this is something which very early in the debate with Heisenberg, for instance, gets pointed out — this spreading problem, so that in this sense, while the problem may be of a metaphysical type there are non-metaphysical issues that it presents that are picked up pretty early by some people.

Dirac:

I did not concern myself with this question of the wave packets at that time.

Kuhn:

You know Heisenberg and Born in particular both jumped very hard, and in Born's case sometimes quite intemperately on the sort of thing that Schrödinger is saying about the wave functions whether the wave function itself is or is not — I'm not sure to what extent this happened before the recognition of the identity of the two. Now I wondered whether that sort of issue had involved you at all at the start or whether you had reacted —.

Dirac:

No, it didn't. I was quite happy just to go my own way and let the other people go their

way with their theories.

Kuhn:

In a number of places one finds that to people who had paid very little attention to matrix mechanics suddenly quantum mechanics begins to seem all right with the wave equation. Was there any of that also at Cambridge?

Dirac:

I don't remember. It might possibly have applied to Darwin, but I'm not sure about that. It's a great pity that Darwin died just a short time ago. But Fowler was quite happy with the matrix mechanics.

Kuhn:

You wouldn't say that with the coming of wave mechanics more people at Cambridge began to get really seriously involved than had been on their way to doing this before.

Dirac:

I think the experimental people were a good deal happier with the waves • That was something they could understand.

Kuhn:

When you began to think that perhaps the q-number approach and the wave function approach were equivalent, did you attempt yourself to find an equivalence between them?

Dirac:

Other people did it first. I heard that other people had established an equivalence between matrix mechanics and wave mechanics and I just read their work.

Kuhn:

You hadn't tried it yourself?

Dirac:

I hadn't tried it, no.

Kuhn:

By the time you heard that they had done it, do you suppose you were pretty well prepared for that or had the idea really —?

Dirac:

I think I was prepared for it when it came.

Kuhn:

You yourself first produced the Schrödinger equation in the paper on the theory of quantum mechanics. [Paper 14] You generalize it by producing a time dependent form which is linear and also by giving, as you produce it, a number of hints of things that are going to show up as transformation theory a little bit later. You talk about j —.

Dirac:

The linear equation that you are referring to is not the equation of the electron is it?

Kuhn:

No, no, no. Again, the timing is hard to tell because the two papers are too nearly at the same time. Schrödinger's early equations, when they've been in a time-dependent form, have involved the second partial with respect to time. He corrects this sometime around mid-summer of '26, which is around the same time that your paper on the theory of quantum mechanics goes in. You both get at this point a form in which only the first derivative with respect to time occurs. These are quite independent? Is that right?

Dirac:

I expect what happened was that after people had established the equivalence between the matrix and the wave theories, I just studied their work and worked on it and tried to improve it in a way that I had done several times previously. I think the transformation theory came out from that.

Kuhn:

The transformation theory doesn't in any very clear-cut form exist in the paper I mean to be pointing to at the moment. That's a slightly later paper. This is a paper that you submit in August, 1926, called "On the Theory of Quantum Mechanics," [Paper No. 14] and the paper that clearly does have transformation theory is one that goes in in December called the "Physical Interpretation of the Quantum Dynamics." [Paper No. 16]

Dirac:

Well, I better check up on this one on the theory.

Kuhn:

This is the one that's now perhaps best-known because it's got the Fermi-Dirac statistics. In fact, it's got an awful lot of things, not obviously parts of the same problem although

Dirac:

By the way, do you know this story that I wrote about Schrödinger?

Kuhn:

No, I don't know this story, and I certainly should.

Dirac:

About the discovery —

Kuhn:

I have read the obituary that you did of Schrödinger.

Dirac:

Well, this story here, I've published it before.

Kuhn:

I know why I don't know this, my copies of the Scientific American the May American had not arrived when I left Copenhagen. [Looking at the article] Yes, this is terribly interesting; you also tell it in the obituary, which you pointed out to us before, so I did know this.

Dirac:

The editor asked me for more details about this and I tried to reconstruct the equation. This is what I imagined the equation to be but I presume that his first equation was this one. He didn't actually write it down to me but I think it must have been that, because it was a generalization of de Broglie's equation and a relativistic one.

Kuhn:

Excellent. I wondered a good deal myself what it would have looked like. This is it, yes. You immediately go to this form in which minus W is minus in partial with respect to t which is not Schrödinger's equation at that point. Actually this is not for you a great big step because it fits so exactly with the W, t formulation that you've given in the Compton effect paper • Without that background it would have been at least a much larger —.

Dirac:

I presumed that Schrödinger had this right from the beginning

Kuhn:

No, no, no.

Dirac:

Well, if he had this equation, then he had it.

Kuhn:

Ah, Ok. But then let me only suggest that that may be an argument against his having had this equation, because there's no question that in the early non-relativistic papers up until part three, until the dritte Mitteilung, when he writes down a time dependent equations at all, he writes it as a wave equation with a second partial, with respect to t and this would have to give him a in it.

Dirac:

Well, you mean he would have this equation without putting d/dt equal to W .

Kuhn:

Yes.

Dirac:

Yes, I see, all right.

Kuhn:

In the dritte Mitteilung, which is actually the fourth paper in the series, he then does correct this time dependent form and comes out more or less at the same time you do with the form that's linear in time. But then you go on here in this to talk about the solutions of the linear equation $FP = 0$ [reading]: "the matrix representation we have obtained is not unique since any set of independent eigen functions will do." Now here, as soon as you start to talk about the characteristics of different sets and then producing one set that will diagonalize W one's taken an important step toward transformation theory although clearly the thing stops right there in this paper. There's certainly nothing like the ψ function as a transformation function. This leaves me wanting to ask whether this brings back any of the ways in which this set of ideas fits in with the development of the fuller transformation theory. I'm terribly interested to know how you were driven toward the fuller transformation theory. I think that paper is of almost all of them to me the most exciting.

Dirac:

The transformation theory one?

Kuhn:

This is not meant to be a historian's evaluation of your work, but that paper I find in particular extraordinarily exciting to read • But there are elements of it here.

Dirac:

I can't remember how I came to think of it. I know it was a question of working on it for a good long time and gradually changing from one form to a slightly better form. I didn't think of it in the first place in the form in which it was published.

Kuhn:

Bohr, in a paper that he did for the Heisenberg Volume, makes the remark that you had said when you heard Heisenberg's report on colloquium in Copenhagen on a paper on fluctuations phenomena that you suddenly saw it was a transformation theory or something of this sort. This story doesn't quite make sense to me, but it does suggest that there may have been something quite important about this colloquium report of Heisenberg's in the development of your transformation theory paper. [Paper No. 16] Do you remember that? It was the report in which Heisenberg points out that the time averages of energy, energy squared, and so forth in a resonant system come out in such a way that one must have a discontinuous passage from one energy level to the other, that one can't have a traceable oscillation in time. You yourself, in the transformation theory paper, very early mention this still unpublished work.

Dirac:

I think that Colloquium report came while I was working on the subject.

Kuhn:

Yes, the date would come after this paper, but before the —

Dirac:

I think I did appreciate it that it fitted in with my work.

Kuhn:

But you don't remember having had a sudden flash of illumination in the course of contemplating this result?

Dirac:

No, no. I think I just felt that it did fit in a way that things ought to fit.

Kuhn:

Yes. This was your own reference to the —.

Dirac:

Yes. I think the answer to that is that I did not get a sudden flash at that time. I don't think I could get a sudden flash from a conversation. I could only get it by turning things over and over.

Kuhn:

Well, if I remember Bohr's words in that paper they'd not be incompatible with your having gone home with the idea and turning it over and over and over and then seeing something in it.

Dirac:

I think I'd already gone a good way towards that idea when I heard Heisenberg talk on it.

Kuhn:

Coming back to this one. You go immediately -.

Dirac:

Do you know the date of Heisenberg's talk?

Kuhn:

Not exactly. It would have been in that fall, but whether early or late in the fall I've got no notion. You go immediately from the generalization of the Schrödinger equation to using it on the problem of similar particles. It works very nicely; on the other hand, just as a line of development in the paper this seems like quite another problem. I wondered how the two had gotten together. You know the problem you solve here is just the problem that Heisenberg at substantially the same time solves, officially using matrix mechanics. It isn't totally clear to me that he didn't get some of that by utilizing wave equations and then reformulating it in a matrix fashion. I wondered how this problem itself developed but wondered particularly about the thing that emerges here that this is the point where you feel you have to throw away the formulation in uniformizing variables which had been the standard q-number interpretation up to this point. [See paper 14: "On the Theory of Quantum Mechanics," Proc. Roy. Soc. A 112 (1926) paragraph 3)

Dirac:

Well, this piece of work was separate from the other one and the only reason why I didn't put it in a separate paper was that I thought the papers would be too short. I was changing over to the wave function formalism because it was more powerful than just the q-numbers.

Kuhn:

When you speak of this as separate do you have any notion of the order, for example, in which they came? To reach this conclusion about the inadequacy of the uniformizing variables you don't need to have the Schrodinger equation. And in a sense your whole experience trying to use action and angle variables for multi-electron problems may have led you to suppose that they wouldn't work for a two electron problem with q-numbers either.

Dirac:

I think they would work apart from the symmetry questions. Wait a minute, they work only for multiply periodic system, don't they?

Kuhn:

I think this is the only way in which they have been developed.

Dirac:

Oh, yes, I should have said that. That's wrong, they work only for multiply periodic systems so they won't work in any case for helium.

Kuhn:

This is exactly the sort of question I mean to be asking. You produce an argument based on symmetry to show that the approach through uniformizing variables breaks down. Now in a sense —.

Dirac:

Did I actually say that?

Kuhn:

Yes, I think so. I think the first place you say it is, [paper 14, p.668) "It may be shown, however, that there is no set of uniformizing variables for a system containing more than one electron." Now, there's one sense in which you must have known that from the start because you'd had that trouble with the old Hamilton-Jacobi techniques. But there's another sense in which you seem to have been expecting it to work in fact surely you had because there's old paper on the nodes in quantum mechanics [Paper No. 10) which deals with multi-electron systems and uses uniformizing variables— and I wondered at what point this conclusion came in.

Dirac:

I better check out this reference here where I said -. Because this second sentence seems to be a doubtful one. I wouldn't expect that —. [searching for paper] ... [examining paper) I've forgotten the exact sense in which I'm using uniformizing variables.

Kuhn:

Well, I have been reading them as meaning the same thing as action and angle variables, that is, as a set of variables such that at least classically the I think they are indistinguishable from a set of action-angle variables. The H is a function only of the momentum variables. The q 's, the original q 's and p 's are multiply periodic. [see, for example] paper No. 9, paragraph 4.)

Dirac:

Well, in that case it would not work for a two electron problem even if the electrons were not equivalent and symmetry conditions don't arise.

Kuhn:

Well and good, but there is the paper [Number 10] that immediately follows this one in which I think the same approach is utilized and one proceeds ahead to multi-electron problems. Now I May be wrong.

Dirac:

This is treating it only by an approximation which does allow uniformizing — action and angle variables. I think it's essentially a one-electron problem with a core like one worked on in the Bohr-Sommerfeld quantization. People worked on the spectra of the alkali metals and things like that and they used action and. angle variables but they were just in an approximate model.

Kuhn:

No, I do see that and I hadn't seen the sense in which this approach would, of course, justify it.

Dirac:

This paper [Number 10] on the nodes involves the same approximation as the old work of Bohr-Sommerfeld.

Kuhn:

I'd realized all along that it did, but I had not quite seen the relation of that to the question of the legitimacy of the uniformizing variables in the multi-electron problem, that they came back in with this approximation. This still leaves me with trying to see whether we can find out any more about the source and sense of this announcement that uniformizing variables won't work with more than one electron. This makes this whole point somewhat stranger because in one sense one wants to say that you knew all along, that they wouldn't work because it's no longer a multiply periodic problem, but on the other hand pretty clearly this proof makes no reference to that; it's based on symmetry properties and it does sound somehow or other as though this is a new idea to you and somewhat discouraging in terms of what it has to say about the previous approach. Or how fundamental the previous approach is.

Dirac:

Of course, at that time one didn't know how far the matrix mechanics would go towards solving problems like the helium spectrum. It could be that one supposed that one would have action and angle variables occurring even there because the whole thing was based on just a limited number of frequencies occurring, corresponding to the different matrix transitions.

Kuhn:

I would think that that was very likely the way one had felt about it, simply because there is no reference to an expectation that this whole thing's going to break down as soon as we get to two-electron problems, and because the proof that it breaks down, when it comes, is of such a different nature. It's a proof based on the symmetry argument. Do you know how you came to trying the symmetry argument and to realizing it's consequences?

Dirac:

I don't think there was any difficulty about coming to the symmetry argument as soon as one faces up to the question that the two-electrons are equivalent. In fact, when two people do a piece of work simultaneously there's really no difficulty about the approach to this work.

Kuhn:

It at least indicates that somehow or other it was in the air but it doesn't mean that the two individuals came to it the same way.

Dirac:

Well, it means that it was pretty obvious. Things which are not pretty obvious are those which take many years before there's a breakthrough. I don't think it's any puzzle how someone gets the idea which leads to the break.

Kuhn:

Yet I refuse to take it that because two people got it, it must have been easy. I will apply this in most cases but when the two people are yourself and Heisenberg

Dirac:

I think it was easy and that any good research student would have got onto it pretty soon.

Kuhn:

Well, if anybody had said to him, “Figure out what the consequences of the indistinguishability of the electrons are.” But what was it like to pick up that idea?

Dirac:

I think people knew for a long time that the electrons were indistinguishable one from another.

Kuhn:

Yes, but I don’t think they’d been using it this way.

Dirac:

Of course, as long as you have electrons moving continuously, it doesn’t bring in any new situations. It’s only when you have jumps that you have new situations coming in. I suppose the only point to appreciate is that this bringing in of jumps does bring in some new and quite indistinguishable particles.) It’s a sort of like the negative energy states: it’s only when the jumps come in that they become important.

Kuhn:

But you’ve got no recollection of how the notion of asking that set of questions came about.

Dirac:

No, no. In any case it was pretty straightforward.

Kuhn:

With the case of the problem of transitions to negative energy levels it’s pretty clear that if you got the equation with the negative energy levels you’d have to ask yourself, Can I avoid transitions?” and then answer, “No.” There’s nothing here that I’m aware of that makes it pressing for you to ask the question you’re here asking.

Dirac:

Well, I think there is. If you just talk about transitions, ‘why do two electrons change places?’ And try to fit such transitions into the theory. You are pressed in that way.

Kuhn:

Yes. Was it hard to drop the idea of the uniformizing variables as a general approach?

Dirac:

Not when one has something else to replace it, namely the Schrödinger wave functions. One doesn't mind dropping an idea if one has another powerful one to use in its place.

Kuhn:

Then you go on right from this to the statistics.

Dirac:

Yes. Again, because it simply made too short a paper if it were put separately.

Kuhn:

But this does fit with the statistics pretty well.

Dirac:

Yes, yes.

Kuhn:

Because it leads you immediately to the symmetric and anti-symmetric —. Had you been perplexed about a reason for the exclusion principle before you got to this? Was that a problem that had—?

Dirac:

It hadn't worried me. If a theory is advancing in one place, one isn't bothered by a temporary stagnation in another place.

Kuhn:

Of course, the exclusion principle exists even before the theory.

Dirac:

Yes. Before the beginning of quantum theory.

Kuhn:

Yes. It was rather the same thing you say about spin. “But why should it be that way” could be, and I think it was, asked in some places about the exclusion principle. “Why shouldn’t two electrons .j Now here it suddenly drops out but that had not been a source of vexation for you — the apparently ad hoc nature of the exclusion principle.

Dirac:

No, there were so many things one didn’t understand. One of the main things was the difference between positive and negative electricity which seemed to be something fundamental in Nature.

Kuhn:

People did not think it was enough to say, “Look, there are two types of particles and one of them’ s large and positive and one of them’ s small and negative.”

Dirac:

Well, people did accept that but it did mean there was a difference between positive and negative electricity. There was not symmetry between them although the electromagnetic equations were symmetrical.

Kuhn:

Who worried about that? Do you remember?

Dirac:

I don’t think “worried” would be the right word to use, One just accepted it as a principle of Nature.

Kuhn:

Well, there’s a terribly big difference in attitude between accepting something as a principle of Nature and feeling that, “All right, yes, it’s a principle of Nature but darn it, everything else I know about Nature makes this an uncomfortable principle because I expect symmetry; there ought to be more to be said.”

Dirac:

Well, there was so little known about atomic theory in those days.

Kuhn:

But when you start out by saying to me that there is another question like the question of why the exclusion principle was, namely, “Why this asymmetry?” that does somehow or other mean that although people may have been sure that the asymmetry was there, they were bothered by it, that it seemed unnatural or that it seemed to call for more explanation.

Dirac:

I don’t believe that’s so. I think people accepted it as the natural order of things and that’s why it was so hard to think of the positron.

Kuhn:

My impression was that people simply had accepted it and hadn’t asked, “Why the asymmetry?” but I think people did ask, “Why the exclusion principle?”

Dirac:

Yes, I don’t remember asking it myself. I accepted it in the same way as the asymmetry.

Kuhn:

No, but you did ask, “Why spin?”

Dirac:

That was something which was much more recent and not so well-established and got thrown at one as a new idea.

Kuhn:

Is there any truth to the rumor that you made a bet with Heisenberg on the subject that you would find a more fundamental explanation than spin?

Dirac:

I think it’s unlikely. I don’t think I would make a bet on the future development of science because it is so uncertain. Who said that? Did Heisenberg say that?

Kuhn:

No, I don't think he did. I've heard it twice, both times I think with appropriate qualifications of the possible fallibility of memory and they are also not necessarily independent. One of them was from Oskar Klein, the other one from somebody else in Copenhagen, I think probably not from Bohr, but probably from somebody Bohr had told it to.

Dirac:

I might have said that I was dissatisfied with the theory and that I believed that someday a better theory would be discovered.

Kuhn:

No, we have a version which I will not vouch for at all which says that you were working on it and bet that you would have the answer within n months where n is some not overwhelmingly large number.

Dirac:

I don't think I've ever had that much confidence when I'm working on a problem. But I might have said that someday someone would find the answer. Just like now I'll say that someday someone will find the reason for the constant.

Kuhn:

You feel that that has not already been answered.

Dirac:

No. I don't think anyone claims to have answered it. Except perhaps Eddington.

Kuhn:

Well, it was Eddington that I had in mind in asking.

Dirac:

We're really now in Copenhagen and I should ask you why you went in the first place but I gather that for a student of in quantum mechanics, wanting to go to Copenhagen was pretty natural. Were there any problems about arranging it? D Fowler was keen that I should go to Copenhagen and I think he wanted me to spend the year there. I was a bit worried about going to a country where I didn't know the language and. I was rather

more desirous to go to Göttingen for that reason, because I did know a little German. We made a sort of compromise between those two by going half a year to Copenhagen and half a year to Göttingen. I had no financial problems then because I had been awarded this studentship from the exhibition of 1851. Did I tell you about that?

Kuhn:

You came to St. Johns originally on an exhibition scholarship.

Dirac:

With an exhibition from the college and with a government grant. But that involved only a small amount of money and I was short of money. Two years later I was awarded another exhibition on a foundation which comes from the profits of the international exhibition that was founded in 1851. Then I had much more money and I had no trouble about financing the traveling.

Kuhn:

Did Copenhagen seem very different as a place to do physics. Different from Cambridge?

Dirac:

I had quite a different life there.

Kuhn:

In what way?

Dirac:

I was doing most of my work there in the institute instead of in my own room or in the library. I think I met people more often there.

Kuhn:

Did you work more with people or was the actual work still done very much by yourself?

Dirac:

It was done by myself. I've never been good at collaborating.

Kuhn:

Many of the people who went to Copenhagen from other places speak of the difference in the approach there or contrasts with the places they had been before. It was different from Göttingen, it was different from Munich; these are the contrasts I hear most of.

Dirac:

Well, there was the personality of Bohr. That was the big thing there. Without Bohr I think there would have been nothing. I was very much impressed by hearing Bohr talk.

Kuhn:

What particularly impressed you about that?

Dirac:

Well, he's a deep thinker and he does think about all problems. -

Kuhn:

You said before that you wouldn't be concerned with problems which are meta physical; did you ever feel that way about any of the problems that Bohr presented?

Dirac:

No, he did not concern himself with metaphysical problems. But he concerned himself with problems which are not related to science at all. For instance, when two gunmen each draw a pistol and each point it at the other, each one wants to kill the other one, but no one dares to shoot. What's the explanation for that? Why doesn't one of them shoot? Well, Bohr worked that out. You know that story? Well, it's a psychological question and if you make up your mind to shoot and then shoot, that's a slower process than if you shoot in response to some external stimulus. If you make up your mind to shoot, the other man will see that you've made up your mind to shoot before you've shot and he will shoot you first. Bohr bought some toy pistols and tried this out with various people in the lab.

Kuhn:

This is a problem presented by his having gone to a western?

Dirac:

I don't know how he thought of it in the first place, but they say that the gunmen know

this. Neither of them dares to shoot because he will be dead before he can carry out his resolution if he does make a resolution to shoot.

Kuhn:

That's fascinating and it is a Bohr story I have certainly not heard before.

Dirac:

It is rather a slow process to make up your mind to do something and then to act on what you've made up your mind to do. It takes perhaps half a second or something like that while it's a much smaller fraction of a second to respond to an external stimulus. Another thing he thought out was about the stock exchange. Someone who studies what stocks to buy and sell and just buys when he thinks he's got something good to buy and sells when he thinks it's going to go down, such a person will do worse than someone who buys and sells completely at random. Bohr worked that out also. And do you know the reason for that? Did you hear that story?

Kuhn:

No, I haven't heard this story. Tell me the reason for it.

Dirac:

The reason is that there are some people with inside information, the directors of the companies and their friends and they are able to make a profit from their inside information; they may see that the company's doing badly. So they sell their shares promptly or they may see that the company's going to do well from confidential information they've received, so they buy the shares. They're able to make a profit. That's obvious. Now if someone buys and sells completely at random, on the whole his profits and losses will cancel each other, assuming there's no fraud, and that he deals with standard shares. The person with inside information cannot get money from the person who buys and sells at random. How can he get money? Well, he gets money from the person who just studies the situation and has a little information. For instance, the director who knows that the company is doing badly and proceeds to sell his shares, in order to get rid of his shares he has to sell them somewhat below their apparent value. The person who just studies the market without inside information sees some shares going for a lower price than their apparent value says, "Well, now here's something good to buy." And he's the one who buys it and loses.

Kuhn:

This one I find unconvincing but I will have to work on it. ...

Dirac:

If too many people knew this Bohr theory and acted on it then the people with inside information would have to work the other way. It's the sort of thought he liked to work with, and he impressed me very much with the extent of his activities.

Kuhn:

What was he working on in quantum mechanics while you were there?

Dirac:

I can't remember what Bohr himself was working on.

Kuhn:

Did you talk much to him about your own work as it went on?

Dirac:

I think mostly Bohr was talking and I was listening. That rather suits me because I'm not very fond of talking.

Kuhn:

Did you give a colloquium on the transformation theory?

Dirac:

I'm pretty sure I did, yes. I did give colloquia from time to time. I can't remember in detail which ones were given. I don't suppose anyone has kept a record.

Kuhn:

I don't think there's anything like a book on them there. There are remarks about some of them from time to time, but I think there's no running record of the colloquia. Do you remember any particular colloquia you went to there?

Dirac:

I can remember Ehrenfest being present sometimes and what a useful person he was to have at a colloquium.

Kuhn:

That was your first contact with him?

Dirac:

Yes.

Kuhn:

He was just as useful in this period perhaps as he had been before?

Dirac:

Oh, yes.

Kuhn:

Because I think he himself felt that he'd never gotten the bang of matrix and wave mechanics.

Dirac:

He was very useful in getting things cleared up in the colloquia where things were not properly presented by the speaker. Ehrenfest was the most useful man that one has ever had at colloquia, I should say.

Kuhn:

Did he do this to Bohr also?

Dirac:

I don't think he did. I can't remember any example. I'm not sure.

Kuhn:

In some way they impress me as having been a good deal alike; but I think not in this way.

Dirac:

Yes, there was some symmetry between them. I suppose you know the remark that Bohr

made so often, “Nicht um zu kritisieren, nur um zu lernen;” when cross questioning someone about the work, and not wishing to offend them, he would put in that remark.

Kuhn:

Did you see much of Heisenberg there?

Dirac:

I don't think he was there very much during the period I was there. But one can look that up.

Kuhn:

I would have thought that he was there that entire fall except perhaps for brief trips out.

Dirac:

I don't remember.

Kuhn:

Did you see much of Klein?

Dirac:

Yes, yes, I think so. But I was in Copenhagen more than once and I may be confusing the two times. Gamow was there one time. I'm not sure whether he was there at that time,

Kuhn:

I don't think he was there yet then. This was the period when the interpretation problem was beginning to loom quite large.

Dirac:

Yes.

Kuhn:

Certainly for you, certainly for Heisenberg, and for Bohr. Do you remember at all what was going on with respect to that problem?

Dirac:

I was mainly absorbed in developing my own ideas. I was content to carry on that way and have other people develop their ideas.

Kuhn:

I realize I've kept you at this longer than I should and it seems to me an excellent place to stop for today.